VisionFood QAI: A Deep Learning Based Quality Inspection System for Food and Beverage Products

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ABSTRACT Quality inspection in the food and beverage industry remains predominantly reliant on manual visual assessment, which suffers from subjectivity, operator fatigue, and limited scalability on high-speed production lines. This paper presents VisionFood QAI, a deep learning based quality inspection system for food and beverage products, encompassing a comprehensive literature survey, detailed implementation methodology, and experimental evaluation. We systematically review 16 recent studies spanning convolutional neural network (CNN) based classification, You Only Look Once (YOLO) based object detection, Internet of Things (IoT) and sensor fusion methods, and specialized techniques including autoencoders and incremental learning. The review identifies five critical research gaps: (1) only 4% of machine learning based food quality control studies target beverages, (2) most systems address only a single defect type, (3) no existing work integrates localization, explainability, remediation, and operator-facing traceability in a unified pipeline, (4) high-accuracy models remain difficult to deploy at the edge, and (5) labeled datasets for food and beverage defects are scarce. Based on this analysis, we design and implement VisionFood QAI with a three-module pipeline: (1) bottle localization and cap defect detection using YOLOv11s, achieving mAP@0.5 of 0.995; (2) cap quality classification using MobileNetV3-Small with Focal Loss, achieving 86.54% accuracy and F1-score of 0.87; and (3) fill-level estimation using a YOLOv8 water surface detector (mAP@0.5 of 0.915) combined with mathematical fill-ratio computation. All models are lightweight variants targeting edge deployment. The system employs a two-pass zoom strategy for small-object cap detection, Monte Carlo Dropout for uncertainty quantification, and supports ONNX/TensorRT export. The product-profile-driven architecture supports transparent bottles, rigid cans, flexible wrappers, and rigid boxes through configurable SKU profiles, integrating explainable AI visualizations, a REMEDY severity-and-triage engine, and a web-based operator dashboard for auditable inspection outputs.

INDEX TERMS Computer vision, deep learning, defect detection, edge inference, explainable AI, fill-level estimation, food quality inspection, MobileNetV3, object detection, REMEDY engine, YOLOv11.

I. INTRODUCTION

THE food and beverage industry is one of the largest manufacturing sectors globally, requiring stringent quality standards to ensure consumer safety and regulatory compliance. Defects such as improper filling levels, packaging damage, label misalignment, and surface contamination directly impact product safety, shelf life, and brand reputation. Traditional quality inspection relies heavily on manual visual assessment by trained operators. However, manual inspection achieves only 70–85% accuracy under fatigue conditions [3], is inherently subjective and inconsistent, and cannot scale to high-speed production lines that generate thousands of products per hour.

Recent advances in deep learning and computer vision have demonstrated remarkable potential for automated, real-time, and highly accurate inspection systems across various manufacturing domains [1], [2]. Convolutional Neural Networks (CNNs) such as ResNet and DenseNet have achieved classification accuracies exceeding 95% on industrial defect detection benchmarks [5], [6], while object detection architectures like YOLO enable real-time defect localization [9], [10]. Despite these advances, a systematic review by [1]

revealed that beverages account for only 4% of machine learning based food quality control studies, highlighting a significant and underexplored research gap.

This paper presents a comprehensive literature survey of deep learning methods applied to food and beverage quality inspection, identifies critical research gaps, and describes the design, implementation, and evaluation of a unified system to address them. The remainder of this paper is organized as follows: Section II presents the problem statement and motivation. Section III provides the detailed literature review. Section IV synthesizes the findings. Section V identifies the research gaps. Section VI describes the proposed system architecture. Section VII details the implementation methodology. Section VIII reports the experimental results. Section IX concludes the paper.

II. PROBLEM STATEMENT AND MOTIVATION

A. PROBLEM STATEMENT

The objective is to design and develop an intelligent, deep learning based quality inspection system that automatically detects, localizes, and analyzes visual defects in food and beverage products, addressing the limitations of manual inspection in terms of accuracy, consistency, speed, and scalability.

Specifically, the system targets five categories of visual defects in packaged food and beverage products:

1) **Fill-level anomalies:** Underfilling, overfilling, and inconsistent volume across a production batch.
2) **Packaging defects:** Cracks, dents, deformation, seal integrity issues, and cap misalignment.
3) **Cap and closure defects:** Missing caps, loose caps, tilted caps, and cross-threaded closures in transparent bottle products.
4) **Label defects:** Missing labels, misaligned or wrinkled labels, and incorrect label information.
5) **Surface contamination:** Foreign particles, residue, discoloration, stains, and product leakage on the exterior.

The system scope encompasses image-based inspection using a YOLO-based object detection pipeline, product-profile-based routing, explainable AI outputs, and REMEDY-based remediation logic. The measurable targets are image-level pass/fail accuracy $\geq 93\%$, defect detection mAP@0.5 $\geq 88\%$, and inference latency < 100 ms on GPU.

B. MOTIVATION

Several factors motivate this work:

- Manual inspection accuracy of only 70–85% under realistic fatigue conditions [3] is insufficient for safety-critical food products.
- High-speed production lines require throughput that manual methods cannot sustain.
- Existing automated systems are often domain-specific and lack generalization across defect types [4].
- The critical underrepresentation of beverages in ML-based food QC research (only 4% of studies) [1] presents a significant opportunity for contribution.
- YOLO-family detectors and ResNet50-based models have demonstrated >95% accuracy in industrial defect detection tasks [5], [6], supporting the use of YOLOv11 for real-time localization and a lightweight MobileNetV3-Small classifier for crop-level decisions in the proposed architecture.

The beneficiaries include manufacturers (reduced waste, recalls, and labor costs), consumers (safer and higher-quality products), and regulators (traceable, data-driven quality assurance).

III. LITERATURE SURVEY

This section reviews 16 recent studies organized into four thematic categories: (A) CNN and Transfer Learning approaches, (B) YOLO and Object Detection methods, (C) IoT and Sensor Fusion systems, and (D) Specialized approaches including autoencoders, incremental learning, and systematic reviews.

A. THEME 1: CNN AND TRANSFER LEARNING

Liquid Filling Quality Control (2024) [7] developed a CNN-based system deployed on NVIDIA Jetson NX for real-time classification of beverage filling levels. The system demonstrated effective real-time inference on edge hardware. However, it addresses only a single defect type (fill-level) and lacks multi-defect capability.

ResNet50 for Defect Detection (2025) [5] applied an optimized ResNet50 architecture with transfer learning on the MVTec Anomaly Detection (MVTec AD) benchmark dataset. The approach achieved high accuracy through fine-tuning pre-trained ImageNet weights. The limitation is the large model size, which poses challenges for edge deployment without quantization or pruning.

Thermoforming Food Package Inspection (2021) [3] conducted a systematic comparison of CNN architectures (DenseNet, ResNet, VGG) for quality inspection of thermoformed food packages. The study confirmed that CNNs significantly outperform manual inspection (70–85% accuracy) but require large labeled datasets for training, which are scarce in the food domain.

Food Oil Quality Detection (2024) [8] proposed a CNN classifier for detecting food oil quality and usage levels from images. The work demonstrated the applicability of AI-based detection to food quality but was limited to a narrow scope (oil quality only) without extension to packaging or manufacturing defects.

Data-Centric AI for Food Packaging (2024) [4] adopted a data-centric AI approach with an end-to-end computer vision pipeline for food packaging quality control. The key insight is that improving data quality yields greater accuracy gains than architectural modifications alone, highlighting the importance of dataset curation over model complexity.

Table 1 summarizes the CNN and transfer learning studies.

TABLE 1. Summary of CNN and Transfer Learning Studies

Study	Method	Dataset	Contribution	Limitation
[7] (2024)	CNN + Jetson NX	Beverage images	Real-time edge classification	Single defect type
[5] (2025)	ResNet50 + TL	MVTec AD	High accuracy via fine-tuning	Large model size
[3] (2021)	DenseNet, ResNet	Food packages	CNN architecture comparison	Needs large datasets
[8] (2024)	CNN classifier	Oil images	AI-based food quality detection	Narrow scope
[4] (2024)	End-to-end CV	Food packaging	Data-centric AI approach	Data > architecture

TABLE 2. Summary of YOLO and Object Detection Studies

Study	Method	Dataset	Contribution	Limitation
[6] (2024)	YOLO + Arduino	Manufacturing	Automated sorting	Arduino limits scale
[9] (2025)	YOLOv8 + PLC	Industrial line	Vision-PLC integration	High hardware cost
[10] (2024)	YOLOv5 review	Manufacturing	YOLO comparison	Speed vs. precision
[2] (2024)	CNN+YOLO+IoT	Smart factory	Complete QC pipeline	Integration complexity
[17] (2025)	YOLOv5+CBAM	Food packaging	5-class defect detection	No XAI; no verdict layer

B. THEME 2: YOLO AND OBJECT DETECTION

YOLO-Based Defect Detection (2024) [6] combined YOLO object detection with Arduino-based sorting for automated defect detection in manufacturing. The system demonstrated end-to-end automation from detection to physical sorting. However, the Arduino-based actuation limits scalability to industrial production speeds.

YOLOv8 with PLC Integration (2025) [9] developed an industrial inspection system integrating YOLOv8 with Programmable Logic Controllers (PLC) for production line deployment. The vision-PLC integration enables direct control of reject mechanisms. The primary limitation is the high hardware cost of the PLC infrastructure.

YOLO for Manufacturing Survey (2024) [10] presented a comprehensive survey of YOLOv5-based approaches for automated defect detection in manufacturing. The survey highlighted the fundamental speed-versus-precision trade-off in YOLO architectures and identified YOLOv8 as the next-generation solution with improved anchor-free detection.

Smart Manufacturing with AI (2024) [2] proposed a complete quality control pipeline combining CNN classification, YOLO detection, and IoT sensor integration for smart factory environments. While comprehensive in scope, the system suffers from significant integration complexity across heterogeneous components.

Food Packaging Defect Detection (2025) [17] developed an improved YOLOv5 algorithm with CBAM attention modules for mechanical automatic food packaging defect detection. The system detects five defect types (scratches, dents, stains, deformations, and label errors) and achieves Precision 0.96, Recall 0.94, and F1-score 0.94 on a 7,400-image dataset. This is the closest prior work to our proposed system. However, it provides no integrated verdict layer, no explainability (XAI), and is limited to wine bottles only.

Table 2 summarizes the YOLO and object detection studies.

TABLE 3. Summary of IoT and Sensor Fusion Studies

Study	Method	Dataset	Contribution	Limitation
[11] (2024)	CNN + IoT	Food samples	Image + sensor QC	Calibration issues
[12] (2025)	RFID + LSTM	Supply chain	Real-time traceability	Cloud latency
[13] (2025)	ML + edge	Manufacturing	Multi-sensor data	Sensor drift

C. THEME 3: IOT AND SENSOR FUSION

AI-Driven IoT for Food Quality (2024) [11] integrated CNN-based image analysis with IoT sensor data for real-time food quality monitoring and analysis. The combination of visual and environmental data provides richer quality assessment. However, sensor calibration across diverse food production environments remains a significant challenge.

RFID and AIoT Platform (2025) [12] developed an RFID multi-sensor system combined with an AIoT cloud platform using LSTM networks for food quality monitoring throughout the supply chain. The system enables real-time traceability from production to retail. Cloud latency introduces delays that may be unacceptable for real-time production line decisions.

Multi-Sensor Smart Manufacturing (2025) [13] proposed an online measurement and quality control system using multi-sensor fusion with machine learning at the edge. The approach leverages complementary sensor modalities for robust quality assessment. Sensor drift over time degrades prediction accuracy and requires periodic recalibration.

Table 3 summarizes the IoT and sensor fusion studies.

D. THEME 4: SPECIALIZED APPROACHES

Machine Vision with LabVIEW (2024) [14] developed a machine vision-based quality inspection system for liquid bottles using LabVIEW with edge detection and pattern matching. The rule-based approach is interpretable and deterministic. However, it lacks the adaptability of learned representations and fails on defects that do not conform to predefined patterns.

Autoencoder-Based Anomaly Detection (2024) [15] applied autoencoder networks for unsupervised anomaly detection in reusable food packaging inspection. The unsupervised approach eliminates the need for labeled defect data. The primary limitation is a high false positive rate, as the reconstruction error threshold must be carefully tuned for each product type.

Incremental Learning for Pharma Bottles (2025) [16] proposed a Learning without Forgetting (LwF) approach with multi-view imaging for automatic visual inspection of pharmaceutical bottles. The system adapts to new defect types without retraining from scratch. However, catastrophic forgetting remains a risk when the distribution of new defects diverges significantly from previously learned ones.

Machine Learning for Food QC: Systematic Review (2025) [1] conducted a PRISMA-based systematic review of over 150 papers on machine learning for food quality control. The critical finding is that beverages account for only 4%

TABLE 4. Summary of Specialized Approach Studies

Study	Method	Dataset	Contribution	Limitation
[14] (2024)	LabVIEW + edge	Bottle images	Pattern matching	Rule-based limits
[15] (2024)	Autoencoder	Food packaging	Unsupervised learning	High false positives
[16] (2025)	LwF + multi-view	Pharma bottles	Adapts to new defects	Forgetting risk
[1] (2025)	PRISMA review	150+ papers	Beverages: 4% gap	Motivates project

of the reviewed studies, revealing a major research gap that directly motivates the present work.

Table 4 summarizes the specialized approach studies.

IV. LITERATURE SYNTHESIS

A. COMPARISON OF APPROACHES

The reviewed literature reveals four major categories of approaches, each with distinct strengths and limitations:

- **CNN-based classification** (ResNet, DenseNet, VGG): These approaches achieve high accuracy (>95%) through transfer learning from ImageNet. However, they are limited to whole-image classification and cannot localize the spatial position of defects within the image [3], [5].
- **YOLO-based detection**: Object detection architectures enable real-time defect localization with bounding box annotations. The fundamental trade-off is between detection speed and localization precision [6], [9], [10]. The recent work by Liu et al. [17] achieves impressive multi-defect detection performance but lacks an integrated verdict, explainability, and remediation layer.
- **IoT and sensor fusion**: Multi-modal approaches combine visual inspection with environmental sensor data, providing richer context for quality assessment. However, they introduce latency, calibration challenges, and infrastructure complexity [11], [12].
- **Autoencoder and unsupervised methods**: These approaches operate without labeled defect data, but suffer from high false positive rates due to the difficulty of setting appropriate anomaly thresholds [15].

B. OBSERVED TRENDS

Four key trends emerge from the literature:

1) A clear shift from rule-based machine vision to **deep learning driven** inspection systems.
2) Growing adoption of **transfer learning** to mitigate the scarcity of labeled food and beverage defect datasets.
3) Increasing emphasis on **edge deployment** and model optimization for real-time inference on resource-constrained hardware.
4) Rising interest in **data-centric AI**, where improving data quality and curation yields greater performance gains than architectural complexity [4].

C. COMMON LIMITATIONS

Across the reviewed studies, the most significant common limitation is the **single-defect focus**: the majority of systems address only one type of defect (e.g., fill-level anomalies OR label errors) and lack a unified multi-defect framework for the food and beverage domain. Furthermore, none of the reviewed systems integrate localization, verdict logic, explainability, remediation, and operator-facing traceability in a single deployable pipeline.

V. RESEARCH GAPS IDENTIFIED

Based on the comprehensive literature review, we identify five critical research gaps:

1) **Limited coverage of food and beverage QC:** The PRISMA systematic review [1] found that only 4% of ML-based food quality control studies target beverages. Broader multi-product food inspection (bottles, cans, pouches, cartons) is even more underrepresented.
2) **Single-defect focus:** Most existing systems detect only one type of defect, lacking a unified multi-defect framework that addresses fill-level, packaging, label, and contamination defects simultaneously.
3) **No decision, XAI, and remediation integration:** The closest prior work [17] achieves multi-defect YOLO detection but provides no integrated PASS/FAIL verdict logic, explainability for operator decisions, or remediation guidance. Food safety standards (ISO 22000, HACCP) require auditable, traceable decisions.
4) **No integrated pipeline:** Few works combine classification, localization, explainability, reporting, and analytics in a single deployable system. Most studies address isolated components.
5) **Edge deployment gap:** Many high-accuracy models (ResNet, DenseNet) are computationally expensive and too large for real-time industrial edge deployment without model compression techniques.

VI. PROPOSED SOLUTION

A. OVERVIEW

To address the identified research gaps, we propose **VisionFood QAI**, a product-profile-driven quality inspection architecture for packaged food and beverage products. The proposed solution combines YOLO-based defect localization with explainability, remediation routing, operator configuration, and dashboard-based traceability. The system is designed not as an isolated model, but as an end-to-end inspection workflow that connects camera acquisition, edge inference, decision logic, data persistence, and human operator interaction.

A central design principle of VisionFood QAI is **SKU-profile-based routing**. In a production line, a single SKU is typically inspected continuously during a shift. Therefore, the operator selects the product and production run once at startup, and the inspection pipeline reads the corresponding product category, sub-type, and thresholds from the SKU profile. This avoids the overhead and risk of per-frame product-type classification while still allowing the same platform to support multiple packaging forms.

B. SYSTEM ARCHITECTURE

The proposed architecture consists of eight tightly coupled stages:

1) **Product and run configuration:** A product is registered with its SKU, product category, product subtype, container contents, expected QR code, expected date fields, and a linked SKU profile. At the start of inspection, the operator starts a production run for one SKU.
2) **Image acquisition:** A standard RGB camera captures product frames from the conveyor line. The architecture assumes controlled lighting and a fixed camera position, avoiding expensive laser or structured-light hardware.
3) **Frame selection and preprocessing:** Candidate frames are selected from the camera stream when the product is stable in the inspection region. Frames are resized, normalized, and converted into the input format required by the detector.
4) **YOLO defect localization:** A YOLOv11s detector performs real-time localization of visible defects and returns class labels, confidence scores, and bounding boxes. The detector is the main visual inspection model and is exportable to ONNX/TensorRT for edge deployment.
5) **Product-specific inspection modules:** The pipeline activates additional checks based on the SKU profile. Transparent bottles enable fill-level, cap-fitting, and contamination checks; rigid cans enable dent and label checks; flexible wrappers enable tear and smudge checks; and rigid boxes enable corner-crush, moisture-stain, and label checks.
6) **Label verification:** QR/barcode verification and OCR-based date verification are applied where configured. This allows the system to identify incorrect labels, missing QR codes, barcode mismatch, and printed date mismatch.
7) **Verdict logic and uncertainty handling:** The system combines detections, confidence thresholds, optional uncertainty estimates, and rule-based overrides to produce PASS, FAIL, ESCALATE, or REVIEW verdicts.
8) **Output, persistence, and dashboard display:** The final inspection result includes detections, annotated images, explainability outputs, severity scores, remediation action, latency, and product metadata. These outputs are persisted and displayed to the operator through the dashboard.

C. PRODUCT PROFILE BASED ROUTING

The proposed architecture uses four product sub-types:

- **Transparent bottle:** PET or glass beverage bottles where the fill level and cap region are visible.
- **Rigid can:** Aluminium or steel cans where structural deformation, label errors, and surface contamination are inspected.
- **Flexible wrapper:** Thin plastic or foil pouches where tears, punctures, smudges, and seal-related defects are relevant.
- **Rigid box:** Cuboidal cardboard packaging where corner crush, panel deformation, moisture staining, and label defects are inspected.

Each SKU profile defines the product category, sub-type, container contents, risk weights, acceptable thresholds, label region, barcode verification settings, OCR date fields, fill-level limits, and contamination thresholds. This makes the inspection logic configurable without retraining or rewriting the pipeline for every product. The profile also separates *packaging shape* from *container contents*; for example, a rigid can may hold either liquid or solid food, and the severity of packaging damage changes accordingly.

D. DETECTION AND DECISION LOGIC

The YOLO detection backbone (YOLOv11s for cap defect detection, YOLOv8 for water surface detection) serves as the main detection model because the system requires real-time localization rather than only image-level classification. Bounding boxes are necessary for operator review, remediation routing, annotated image generation, and defect-specific explainability. The detector produces class-wise confidence scores and normalized coordinates for each detected defect. These detections are then converted into an image-level quality decision using configurable thresholds.

The decision layer follows a priority-based logic. Identity-related label failures such as QR mismatch receive the highest priority because they indicate that the wrong product or batch may be on the line. Fill-level deviations are treated as hard compliance failures when enabled for transparent bottles. Date mismatch is escalated for human review, while ordinary visual defects are processed through confidence and uncertainty thresholds. This allows the system to distinguish confirmed defects from low-confidence cases that should be reviewed rather than automatically rejected.

E. EXPLAINABLE AI AND REMEDY ENGINE

Explainability is treated as a core requirement rather than a visualization add-on. The system generates annotated images with bounding boxes and class-confidence labels for every inspected product. For detections requiring operator justification, detection-specific XAI visualizations are generated using saliency-based heatmaps, such as D-RISE-style explanations, to highlight the image regions responsible for the YOLO model decision. These outputs improve trust, auditability, and operator understanding in safety-sensitive food inspection settings.

The REMEDY engine extends the inspection result beyond pass/fail classification. For each primary defect, it computes a severity score from bounding-box area, confidence or uncertainty, defect-class risk, and number of remediation attempts. The severity grade is then mapped to an action such as RELABEL, REFILL, REPACK, CLEAN, REJECT, or PASS. Product-specific risk weights from the SKU profile

allow the same defect class to have different consequences for different products; for example, packaging damage on a liquid container is more severe than cosmetic damage on a solid-food cardboard box.

F. USER INTERFACE AND CONFIGURATION WORKFLOW

The proposed system includes a web-based operator interface connected to the inspection API. The dashboard is not only a reporting layer; it is part of the proposed inspection architecture because it configures the product context used by the inference pipeline.

The interface supports three main workflows:

1) **Product registration:** Supervisors register a product SKU with its name, category, sub-type, container contents, SKU profile, expected QR code, and expected printed date fields.
2) **Production run setup:** Before inspection begins, the operator starts a production run by selecting the active SKU. From that point, all frames on the line inherit the same product configuration.
3) **Inspection monitoring:** The dashboard displays live inspection status, latest verdicts, defect counts, annotated images, escalation queue, model/device status, analytics, and downloadable reports.

The backend is designed around a FastAPI service layer, with product and production-run metadata stored in MongoDB and inspection history persisted for analytics. The frontend uses a React-based dashboard for product setup, manual inspection, live monitoring, history, escalations, and report generation. This full-stack design ensures that model predictions are linked to product metadata, operator actions, and traceable inspection records.

G. DATASETS

The system is trained and evaluated on:

- **MVTec Anomaly Detection (MVTec AD):** Industry-standard benchmark with 5,354 high-resolution images across 15 object categories including bottles, used for transfer learning baseline.
- **Custom Food and Beverage Dataset:** Product-specific images covering transparent bottles, rigid cans, flexible wrappers, and rigid boxes, annotated according to the proposed defect taxonomy.
- **Public Datasets:** Kaggle Bottle Quality Control, Roboflow Universe food packaging inspection datasets, and bottle-cap datasets for cap-quality detection.

H. TARGET PERFORMANCE

Table 5 presents the quantitative performance targets with justifications. The image-level pass/fail accuracy target of $\geq 93\%$ is based on state-of-the-art CNN and YOLO-family methods achieving high accuracy on industrial defect benchmarks [5], [6], with a conservative margin for food and beverage domain variability. The detection mAP@0.5 target of $\geq 88\%$ is derived from the closest prior work [17], which achieved Precision 0.96, Recall 0.94, and F1-score 0.94, setting a competitive yet achievable baseline. The system-level latency target includes detector inference, routing, verdict generation, and result packaging.

TABLE 5. Target Performance Benchmarks

Metric	Target	Justification
Accuracy	$\geq 93\%$	Conservative target vs. SOTA (95%)
Precision	$\geq 93\%$	Minimize false alarms
Recall	$\geq 97\%$	Safety-critical
F1-Score	$\geq 95\%$	Balanced performance
mAP@0.5	$\geq 88\%$	Competitive with Liu et al. (2025)
mAP@0.5:0.95	$\geq 75\%$	Strict quality
Inference (GPU)	< 100 ms	Real-time capable
Inference (CPU)	< 200 ms	Production-ready

I. TOOLS AND TECHNOLOGIES

The proposed toolchain uses Python, PyTorch, Ultralytics (YOLOv11 and YOLOv8), MobileNetV3-Small for lightweight crop-level classification, OpenCV, Albumentations, and scikit-learn for vision model development and preprocessing. Model export is planned through ONNX and TensorRT for edge-ready inference. The application layer uses FastAPI for inspection APIs, React with Vite for the dashboard, MongoDB for product and production-run metadata, SQLite for lightweight inspection history during local deployment, and Redis/WebSocket support for live dashboard updates where required.

VII. IMPLEMENTATION METHODOLOGY

The VisionFood QAI system implements three core inspection modules in a cascaded pipeline. Each module is designed with edge deployment as a primary constraint, favoring lightweight architectures that maintain real-time inference capability on resource-constrained hardware such as NVIDIA Jetson Nano or Raspberry Pi.

A. MODULE 1: BOTTLE LOCALIZATION AND CAP DEFECT DETECTION

The first module employs YOLOv11s (Small variant, ∼9.4M parameters, 21.5 GFLOPs) for real-time detection and localization of bottles and cap defects. YOLOv11 introduces C3k2 blocks (Cross Stage Partial with two convolution kernels) and SPPF (Spatial Pyramid Pooling – Fast) for improved multi-scale feature extraction. The architecture consists of a CSPDarknet backbone, a Feature Pyramid Network (FPN) + Path Aggregation Network (PAN) neck for multi-scale feature fusion, and a decoupled detection head that separately predicts objectness, class probabilities, and bounding box coordinates in an anchor-free manner.

1) Three-Class Detection

The detector is trained on a custom dataset with three classes:

TABLE 6. YOLOv11s Detection Training Configuration

Parameter	Value
Pretrained weights	yolo11s.pt (COCO)
Image size	736×736
Optimizer	AdamW
Initial learning rate	0.001
LR schedule	Cosine annealing (lrf = 0.01)
Batch size	16
Epochs (max)	100
Early stopping patience	20 epochs
Warmup epochs	5
Classification loss weight	1.5
Mosaic augmentation	0.5
Mixup augmentation	0.1
Rotation	$\pm 10^\circ$
HSV jitter (H/S/V)	0.015 / 0.7 / 0.4
Horizontal flip	0.5
Vertical flip	0.2
Random erasing	0.4
Multi-scale training	Enabled
Hardware	NVIDIA RTX A5000 (24 GB)
System RAM	128 GB
Mixed precision (AMP)	Enabled

- **goodCap:** Properly seated, undamaged bottle cap.
- **defectCap:** Misaligned, damaged, loose, or improperly sealed cap.
- **noCap:** Bottle with missing cap.

2) Two-Pass Zoom Strategy

A key innovation in the pipeline is the two-pass detection strategy to address small-object detection challenges for bottle caps:

1) **Pass 1 (Full Frame):** The detector runs on the full-resolution image at confidence threshold 0.25 to localize all bottles and visible caps.
2) **Pass 2 (Guided Zoom):** For each detected bottle, the top 35% of the bounding box (the cap region) is cropped with 15% padding, upscaled by $2.5\times$ (up to 960 px), and re-processed at a lower confidence threshold of 0.15 to detect caps missed in Pass 1.

Duplicate detections from the two passes are merged using IoU-based deduplication with a threshold of 0.3, retaining the higher-confidence detection. Pass 2 is skipped when Pass 1 already detects a cap with confidence ≥ 0.70.

3) Training Configuration

Table 6 summarizes the training configuration for the YOLOv11s detector. The model is initialized from COCO-pretrained weights and fine-tuned on the custom beverage cap dataset with heavy online augmentation.

B. MODULE 2: CAP QUALITY CLASSIFICATION

The second module uses MobileNetV3-Small [19] as a lightweight crop-level classifier to provide fine-grained cap quality assessment beyond the YOLO detector's full-frame predictions.

1) Architecture

MobileNetV3-Small employs inverted residual blocks with squeeze-and-excitation (SE) attention modules and h-swish activation functions. With approximately 2.5M parameters and ∼57M multiply-adds per inference, it is well-suited for edge deployment. The classification head replaces the default MobileNetV3 classifier with a custom architecture:

1) Adaptive average pooling $\rightarrow$ 576-dimensional feature vector
2) Dropout ($p = 0.3$) for MC Dropout uncertainty quantification
3) Linear ($576 \rightarrow 256$) + Hardswish activation
4) Dropout ($p = 0.3$)
5) Linear ($256 \rightarrow 3$) output logits

The three output classes are: `good_cap`, `defective_cap`, and `no_cap`.

2) Focal Loss for Class Imbalance

To address class imbalance where good caps significantly outnumber defective samples, the classifier uses Focal Loss [20]:

$$\mathcal{L}_{\text{FL}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

where p_t is the predicted probability for the correct class, $\gamma = 2.0$ focuses learning on hard examples by down-weighting easy samples, and class-specific weights $\alpha = [0.25, 0.50, 0.25]$ emphasize the defective cap class. Label smoothing of 0.05 is applied to improve probability calibration.

3) Preprocessing and Augmentation

Input images are preprocessed with aspect-ratio-preserving letterbox resize to 224×224 with gray (114, 114, 114) padding and ImageNet normalization. Training augmentation includes aggressive color jitter (hue $= 0.5$, saturation $= 0.8$), random grayscale conversion (30%), simulated motion blur, specular glare simulation, random color shift, Gaussian blur, and random erasing (15%). These domain-specific augmentations force the model to learn structural features of cap defects rather than color-dependent patterns, improving robustness under variable conveyor belt lighting conditions.

4) Uncertainty Quantification

Monte Carlo (MC) Dropout is employed at inference time by keeping dropout layers active during $T = 10$ stochastic forward passes. The prediction uncertainty is estimated as the variance across these predictions. High-uncertainty detections trigger an `ESCALATE` verdict for human review rather than automatic acceptance, improving safety in critical inspection scenarios.

TABLE 7. YOLO Architecture Comparison for Edge Deployment

Model	Params (M)	GFLOPs	Size (MB)	mAP_{50}^{val}	Edge
YOLOv8n	3.2	8.7	6.3	37.3	✓
YOLOv8s	11.2	28.6	22.5	44.9	△
YOLOv11n	2.6	6.5	5.4	39.5	✓
YOLOv11s	**9.4**	**21.5**	**18.4**	**47.0**	✓
YOLOv11m	20.1	68.0	38.8	51.5	×

C. MODULE 3: FILL LEVEL ESTIMATION

Fill-level anomalies (underfilling, overfilling) are critical quality defects in beverage production. The third module implements a hybrid approach combining deep learning detection with rule-based mathematical computation.

1) Water Surface Detection

A dedicated YOLOv8 model is trained to detect the water surface line within bottle crops. The detector identifies the liquid-air boundary as a bounding box, providing the precise vertical position of the water level within the bottle.

2) Fill Ratio Computation

Given a bottle bounding box $[x_1^b, y_1^b, x_2^b, y_2^b]$ and a water surface bounding box $[x_1^w, y_1^w, x_2^w, y_2^w]$, the fill ratio is computed as:

$$y_{\text{water}} = \frac{y_1^w + y_2^w}{2} \tag{2}$$

$$R_{\text{fill}} = \frac{y_2^{b'} - y_{\text{water}}}{y_2^{b'} - y_1^{b'}} \tag{3}$$

where $y_1^{b'} = y_1^b + 0.03 \cdot h_b$ and $y_2^{b'} = y_2^b - 0.02 \cdot h_b$ are corrected bottle boundaries with 3% top and 2% bottom padding trimmed to account for bounding box overshoot. The fill ratio $R_{\text{fill}} \in [0, 1]$ represents the proportion of the bottle that is filled.

3) Verdict Logic

The fill verdict is determined by configurable thresholds:

$$\text{verdict} = \begin{cases} \text{underfill} & \text{if } R_{\text{fill}} < 0.40 \\ \text{normal} & \text{if } 0.40 \leq R_{\text{fill}} \leq 0.85 \\ \text{overfill} & \text{if } R_{\text{fill}} > 0.85 \end{cases} \tag{4}$$

These thresholds are configurable per SKU profile, allowing different fill-level tolerances for different product types.

D. MODEL SELECTION RATIONALE

The selection of nano and small model variants is driven by the edge deployment objective. Table 7 compares YOLO variant candidates, and Table 8 compares classifier architectures considered during design.

YOLOv11s was selected as the optimal balance between detection accuracy (mAP_{50} = 47.0 on COCO) and computational cost (21.5 GFLOPs), offering 25% fewer FLOPs than YOLOv8s while achieving 2.1 points higher mAP_{50}. For the classifier, MobileNetV3-Small was chosen for its minimal parameter count (2.5M), sub-10 MB model size, and ∼25 ms inference latency, enabling deployment on edge devices with hardware acceleration via ONNX Runtime or TensorRT.

TABLE 8. Classifier Architecture Comparison

Model	Params (M)	Size (MB)	Top-1 (%)	Latency (ms)	Edge
ResNet50	25.6	97.8	76.1	∼150	×
EfficientNet-B0	5.3	20.0	77.1	∼120	△
MobileNetV3-S	**2.5**	**9.8**	**67.4**	**∼25**	✓
MobileNetV3-L	5.4	21.1	75.2	∼60	△

TABLE 9. YOLOv11s Cap Defect Detection — Validation Results

Metric	Achieved	Target
Precision	0.869	≥ 0.93
Recall	0.973	≥ 0.97
mAP@0.5	**0.995**	≥ 0.88
mAP@0.5:0.95	0.814	≥ 0.75
Training epochs	66	100 (max)
Convergence epoch	∼46	—

The edge deployment strategy involves three-stage model optimization:

1) **ONNX export:** Platform-independent format with graph optimizations and operator fusion.
2) **FP16 quantization:** Half-precision inference for 2× memory reduction with negligible accuracy loss.
3) **TensorRT compilation:** NVIDIA-specific optimization with layer fusion and INT8 calibration for maximum throughput on Jetson hardware.

VIII. EXPERIMENTAL RESULTS AND DISCUSSION

This section reports the training and validation results for each module of the VisionFood QAI system. All experiments were conducted on an NVIDIA RTX A5000 GPU (24 GB VRAM) with 128 GB system RAM.

A. YOLOV11S CAP DEFECT DETECTION

The YOLOv11s detector was trained on a custom beverage cap dataset with offline color augmentation for 66 epochs (early stopping triggered at epoch 66 with patience of 20, indicating convergence at approximately epoch 46). Table 9 summarizes the validation metrics.

The model achieved a peak mAP@0.5 of 0.995, significantly exceeding the target of $\geq$ 88%. The recall of 0.973 meets the target threshold, indicating strong defect coverage, which is critical for safety-sensitive inspection where missed defects (false negatives) are costlier than false alarms. The precision of 0.869, while below the aggressive target of 0.93, reflects a deliberate trade-off: the elevated classification loss weight ($w_{\text{cls}} = 1.5$) and heavy augmentation prioritize recall over precision.

Table 10 reports the training loss convergence, demonstrating effective learning across all loss components. The classification loss showed the most dramatic improvement (77%

TABLE 10. Training Loss Convergence — YOLOv11s Detector

Loss Component	Epoch 1	Epoch 66	Reduction
Box loss	0.993	0.385	61%
Classification loss	3.375	0.791	77%
DFL loss	1.353	0.934	31%

TABLE 11. MobileNetV3-Small Cap Classifier — Validation Results

Metric	Achieved	Target
Accuracy	**0.865**	≥ 0.93
Precision	0.870	≥ 0.93
Recall	0.870	≥ 0.97
F1-Score	**0.870**	≥ 0.95

TABLE 12. YOLOv8 Water Surface Detection — Validation Results (Epoch 105)

Metric	Achieved	Target
Precision	0.888	≥ 0.93
Recall	0.878	≥ 0.97
F1-Score	0.883	≥ 0.95
mAP@0.5	**0.915**	≥ 0.88
mAP@0.5:0.95	0.591	≥ 0.75
Best epoch	105	—

reduction), confirming that the model learned to discriminate between the three cap-quality classes.

B. MOBILENETV3 CAP QUALITY CLASSIFIER

The MobileNetV3-Small cap quality classifier was trained with Focal Loss ($\gamma = 2.0$) and class-weighted sampling. Table 11 reports the validation results extracted from the best checkpoint.

The classifier achieved 86.54% validation accuracy with balanced precision and recall of 0.870, yielding an F1-score of 0.870. The balanced precision–recall symmetry indicates that Focal Loss and class-weighted sampling successfully mitigated the class imbalance problem. While these results fall below the aggressive target thresholds, the classifier effectively discriminates between cap quality states and serves as a refinement layer over the YOLO detector's coarse-grained predictions. The combined two-stage pipeline (YOLO detection $\rightarrow$ MobileNetV3 classification) produces more reliable verdicts than either model alone.

C. WATER SURFACE DETECTION FOR FILL LEVEL

The YOLOv8 water surface detector was trained for 105 epochs on annotated transparent bottle images. Table 12 reports the best validation results.

The water surface detector achieved mAP@0.5 of 0.915, exceeding the target threshold of 0.88. The mAP@0.5:0.95 of 0.591 indicates that while the model reliably detects the water surface, precise localization at stricter IoU thresholds remains challenging due to the inherent visual ambiguity of liquid surface boundaries in transparent containers with refraction and specular highlights. However, the mathematical fill-ratio computation (Equations 2–3) is robust to moderate bounding box imprecision because it uses the vertical midpoint of the water surface detection rather than a single edge coordinate.

TABLE 13. Consolidated Performance Summary — VisionFood QAI

Module	Metric	Achieved	Target	Status
YOLOv11s Detector	mAP@0.5	0.995	≥ 0.88	✓
YOLOv11s Detector	mAP@0.5:0.95	0.814	≥ 0.75	✓
YOLOv11s Detector	Recall	0.973	≥ 0.97	✓
Cap Classifier	Accuracy	0.865	≥ 0.93	△
Cap Classifier	F1-Score	0.870	≥ 0.95	△
Water Surface	mAP@0.5	0.915	≥ 0.88	✓
Water Surface	mAP@0.5:0.95	0.591	≥ 0.75	△

D. CONSOLIDATED PERFORMANCE SUMMARY

Table 13 provides a consolidated comparison of all modules against their respective target benchmarks.

The YOLOv11s detector meets or exceeds all target benchmarks, with mAP@0.5 of 0.995 representing near-perfect detection performance on the validation set. The water surface detector exceeds the mAP@0.5 target. The cap classifier achieves strong discriminative performance but requires further improvement to meet the aggressive accuracy targets. Planned improvements include hard negative mining with additional defective cap samples, test-time augmentation (TTA), and ensemble strategies combining the YOLO detector's class predictions with the MobileNetV3 classifier's refined outputs.

IX. CONCLUSION

This paper presented VisionFood QAI, a deep learning based quality inspection system for food and beverage products. Through a comprehensive literature survey of 16 recent studies across four thematic categories, we identified five critical research gaps, including the severe underrepresentation of beverages (only 4% of ML-based food QC studies) and the absence of unified systems that integrate localization, explainability, verdict logic, remediation, and operator-facing traceability.

To address these gaps, we designed and implemented a three-module inspection pipeline: (1) a YOLOv11s detector for bottle localization and cap defect detection, achieving mAP@0.5 of 0.995 and recall of 0.973; (2) a MobileNetV3-Small classifier for cap quality assessment with Focal Loss, achieving 86.54% accuracy; and (3) a YOLOv8-based water surface detector combined with mathematical fill-ratio computation for fill-level estimation, achieving mAP@0.5 of 0.915. The system integrates these modules with SKU-profile-based routing, explainable AI outputs, a REMEDY severity engine, and a web-based operator dashboard.

The key contributions include the two-pass zoom strategy for small-object cap detection, the use of Focal Loss with

class-weighted sampling for imbalanced defect data, and the hybrid detection-plus-mathematical-rule approach for fill-level estimation. Future work will focus on improving the cap classifier accuracy through hard negative mining and expanded training data, optimizing inference latency via TensorRT compilation for NVIDIA Jetson deployment, and conducting real-world production line trials across multiple product sub-types.

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